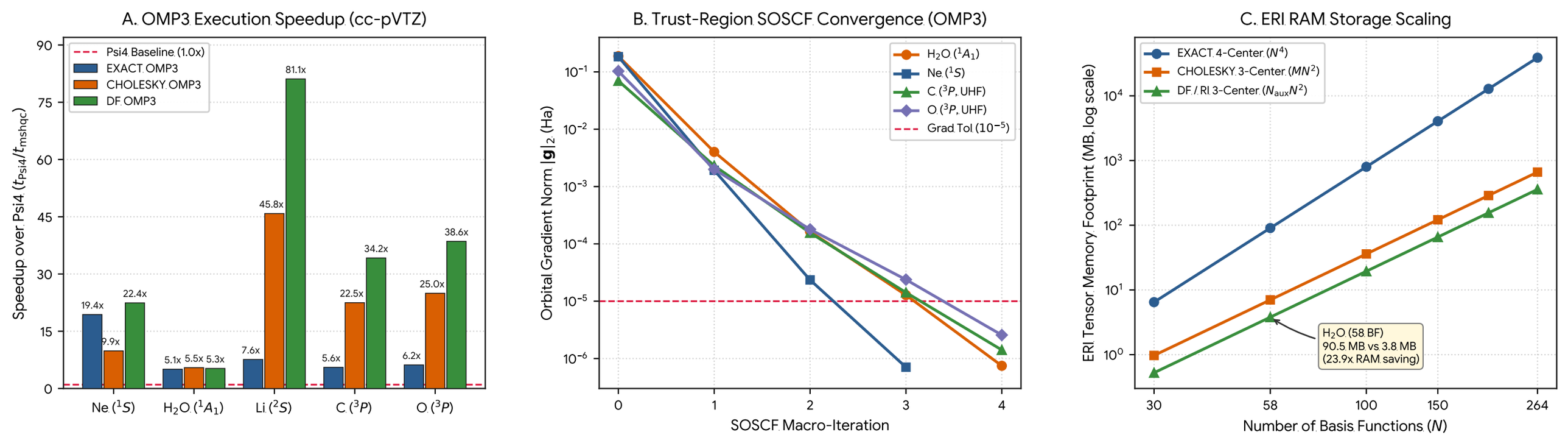
A. OMP3 Execution Speedup (cc-pVTZ)
Psi4 Baseline (1.0x)
EXACT OMP3
CHOLESKY OMP3
DF OMP3
Speedup over Psi4 (t_{Psi4}/t_{mshqc})
0
15
30
45
60
75
90
19.4x
9.9x
22.4x
5.1x
5.5x
5.3x
7.6x
45.8x
81.1x
5.6x
22.5x
34.2x
6.2x
25.0x
38.6x
Ne (^1S)
H_2O ($^1A_1$)
Li (^2S)
C (^3P)
O (^3P)
B. Trust-Region SOSCF Convergence (OMP3)
H_2O ($^1A_1$)
Ne (^1S)
C (^3P, UHF)
O (^3P, UHF)
Grad Tol (10^{-5})
Orbital Gradient Norm $|\mathbf{g}|_2$ (Ha)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
1
2
3
4
SOSCF Macro-Iteration
C. ERI RAM Storage Scaling
EXACT 4-Center (N^4)
CHOLESKY 3-Center (MN^2)
DF / RI 3-Center ($N_{aux}N^2$)
ERI Tensor Memory Footprint (MB, log scale)
10^0
10^1
10^2
10^3
10^4
H_2O (58 BF)
90.5 MB vs 3.8 MB
(23.9x RAM saving)
30
58
100
150
264
Number of Basis Functions (N)